AVID: A Large-Scale, Weakly-Aligned RGB-Infrared Drone and Bird Dataset with a Reproducible Corruption and Alignment-Robustness Benchmark

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Abstract

Vision-based counter-drone systems need to detect and track small, fast, low-signature aerial targets under exactly the conditions that break a single sensor: darkness, fog, glare, and long standoff range. Fusing RGB and thermal infrared (IR) imagery is the natural response, but the public datasets that could support this research have a structural gap: RGB-IR pair datasets (LLVIP, M3FD, RoadScene, KAIST, FLIR ADAS) are pixel-aligned and built around pedestrians and vehicles, while drone-as-target datasets (the Anti-UAV family, CST Anti-UAV) are thermal-only, and drone-*view* datasets (VisDrone, DroneRGBT, VTUAV) point the camera the wrong way. We release AVID (Aerial Visible-Infrared Drone dataset), a large-scale video and image-pair dataset in which drones and a thermally-confusable negative class, birds, are the targets, captured with RGB and IR sensors that are only *weakly* aligned – the realistic case for a dual-sensor gimbal, and a property we quantify per-pair rather than hide. Building on a dataset originally constructed for the IEEE ICIP 2025 VIP Cup (payload-identification data and task excluded from this release by design), we contribute: (1) AVID-C, a reproducible, parameterized extension of the dataset's built-in 7-family corruption taxonomy (speckle, salt-and-pepper, Gaussian blur, uneven illumination, motion blur, camera instability, AWGN) to 5 controlled severities each, in the spirit of ImageNet-C; (2) a per-pair weak-alignment quantification protocol using phase correlation and mutual-information registration; (3) a missing-modality and alignment-sensitivity evaluation protocol that measures how much injected misalignment a fusion model can tolerate before it fails; (4) a documented split-leakage pitfall specific to short video clips, which we show can leak up to 88% of "training" frames into val/test under naive frame-level splitting; and (5) reference detection and fusion baseline code (single-modality, feature-level, and decision-level fusion) released alongside the dataset. We report a real, already-executed cross-modal translation baseline (CycleGAN-Turbo, IR→RGB) with its full training configuration and both its successes and documented failure modes. The dataset is released on Hugging Face under CC-BY-NC-4.0; all benchmark code is released alongside this paper.

1 Introduction

Counter-drone (anti-UAV) vision systems must detect, classify, and track small aerial objects that are adversarial to exactly the sensing modalities that are cheapest to deploy. RGB cameras fail in darkness, fog, and glare, and lose signal entirely once a drone is a few dozen pixels wide at range. Thermal (IR) cameras keep working in the dark and see through some atmospheric scattering, but lack the spatial and textural detail needed to distinguish a drone from a bird, or to read its silhouette at all once its thermal signature is weak (e.g. a coasting, low-power drone against a warm background). The natural response – fusing RGB and IR – is well established for pedestrian and vehicle detection (KAIST [6], LLVIP [7], M3FD [9], RoadScene [16]), but the public data situation for *drones as the fused target* is structurally thinner than it looks at first glance.

The gap. We surveyed the two adjacent lines of public RGB-T and drone-vision datasets (Table 1, Figure 1) and found no dataset that is simultaneously (a) RGB-IR *paired*, (b) has a *drone* (not a pedestrian or vehicle) as the target class, and (c) ships video/tracking data rather than isolated still pairs. The RGB-T pair datasets closest in spirit to fusion research (LLVIP, M3FD, RoadScene, KAIST, FLIR ADAS) are all pedestrian/vehicle detection corpora with no drones at all. The datasets built specifically for anti-drone vision (Anti-UAV [8], Anti-UAV410 [5], the 3rd Anti-UAV Challenge [4], CST Anti-UAV [15]) are thermal-*only*: there is no paired RGB stream to fuse against. And the datasets that do pair a drone-mounted camera with thermal imagery (DroneRGBT [12], VTUAV [17]) or with RGB alone (VisDrone [19]) have the camera on the drone, looking down at pedestrians or generic scenes – the inverse configuration from anti-drone surveillance, where the camera is on the ground looking up or out at the drone.

What we release. AVID closes this gap: RGB-IR paired video and frame-pair data with drones and birds as the target classes, captured under realistic ground-based surveillance conditions (hilly regions, forest cover, cloudy sky, fog/mist), at scale (tens of thousands of pairs; see §3), and – critically – with RGB and IR sensors that are only *weakly* aligned, because that is what a real dual-optic gimbal actually delivers. Rather than pretend the pairs are pixel-registered (as LLVIP/M3FD/KAIST's co-located or heavily calibrated rigs allow), we quantify the residual geometric offset per pair and publish it as metadata, and we use that weak alignment as the starting point for a new empirical question: *how much does RGB-IR alignment precision actually matter for drone detection fusion?* (§5).

Contributions.

1. **AVID dataset**: large-scale, weakly-aligned RGB-IR drone/bird detection and tracking data (video + frame pairs), released on Hugging Face under CC-BY-NC-4.0, with video-level splits and per-pair alignment metadata (§3).
2. **AVID-C corruption benchmark**: a reproducible, parameterized (7 corruption families $\times$ 5 severities) extension of the original dataset's fixed-severity distortions, released as code so any subset can be corrupted at any severity, deterministically (§4).
3. **Weak-alignment quantification and sensitivity protocol**: two independent per-pair alignment estimators (phase correlation on edge maps, and mutual-information registration search), plus a harness for measuring model degradation under controlled injected misalignment (§5).
4. **A documented split-leakage pitfall** for short video-clip corpora, with a reproducible ablation showing naive frame-level splitting can leak the large majority of "held-out" frames (§3.5).
5. **Reference baseline code** for single-modality, feature-level (CBAM-style attention fusion), and decision-level (Weighted Boxes Fusion) detection, plus a documented, already-executed cross-modal translation baseline (§6).

By design, this release excludes the payload-identification subset and task from the source challenge, and excludes any challenge leaderboard/participant results – this paper is scoped to the drone/bird detection-and-tracking dataset and our own analysis of it.

2 Related Work

Table 1 summarizes the datasets closest to AVID along the three axes that matter for this release: whether RGB and IR/thermal are paired, whether a drone is the detection target (rather than the imaging platform), and whether the data includes video/tracking rather than isolated stills. Figure 1 positions them schematically on the first two axes.

Table 1: RGB-IR paired and/or drone-vision datasets closest to AVID. "Aligned" reflects the alignment methodology reported by each dataset's authors, not a re-measurement by us. "Drone target" distinguishes datasets where a drone is the object being detected/tracked from "drone view" datasets where the camera is drone-mounted.

Dataset	RGB+IR paired	Drone target	Video/tracking	Alignment	Scale
LLVIP [7]	✓	–	–	pixel-aligned	16,836 pairs
M3FD [9]	✓	–	–	pixel-aligned	4,200 pairs
RoadScene [16]	✓	–	–	registered	221 pairs
KAIST [6]	✓	–	✓	pixel-aligned	∼95K pairs
FLIR ADAS [14]	✓	–	–	weakly aligned	26,442 frames
Anti-UAV [8]	– (IR only)	✓	✓	n/a	300+ sequences
Anti-UAV410 [5]	– (IR only)	✓	✓	n/a	410 sequences
CST Anti-UAV [15]	– (IR only)	✓	✓	n/a	220 sequences
DroneRGBT [12]	✓	– (drone view)	–	registered	∼3,600 pairs
VTUAV [17]	✓	– (drone view)	✓	well-aligned	500 sequences
VisDrone [19]	– (RGB only)	– (drone view)	✓	n/a	∼10K images
AVID (this work)	✓	✓	✓	**weakly aligned, quantified**	45K/6.5K/13K pairs

RGB-T pedestrian/vehicle fusion. LLVIP [7] and KAIST [6] are the standard low-light pedestrian fusion benchmarks, both using co-located or beam-splitter rigs to achieve pixel alignment; M3FD [9] and RoadScene [16] target vehicle-scene fusion with similarly tight registration. FLIR ADAS is the one widely-used dataset in this group that is *not* pixel-aligned (different optics/FOV between the RGB and thermal cameras), which is closer to AVID's capture setup, but it contains no drones and is a still-image, not video, corpus.

Anti-drone vision. The Anti-UAV line of benchmarks [4, 5, 8, 15] is the closest prior art on the *target* axis – drones are exactly what these datasets track – but every one of them is thermal-only. There is no RGB stream to fuse against, so none of the RGB-T fusion techniques developed for LLVIP/M3FD-style data has a natural anti-drone benchmark to be evaluated on. AVID is, to our knowledge, the first publicly-documented dataset to close this specific gap.

Drone-view datasets. DroneRGBT [12], VTUAV [17], and VisDrone [19] all put the camera on the drone rather than pointing at one. They are valuable for aerial perception broadly, but structurally answer a different question (what does a drone see) from the one AVID targets (what does a ground-based, dual-sensor surveillance system see when it looks at a drone).

Corruption-robustness benchmarking. Our corruption suite follows the severity-graded corruption-family methodology introduced by ImageNet-C [2]. The closest existing robustness benchmark in the drone space, UAV-C [10], targets a different problem: it corrupts UAV-*view* tracking benchmarks (drone-mounted-camera footage), not anti-UAV RGB-T detection of a drone as the target. We are not aware of an existing corruption-robustness benchmark for the Anti-UAV-style drone-as-target setting, which is the gap AVID-C (§4) targets.

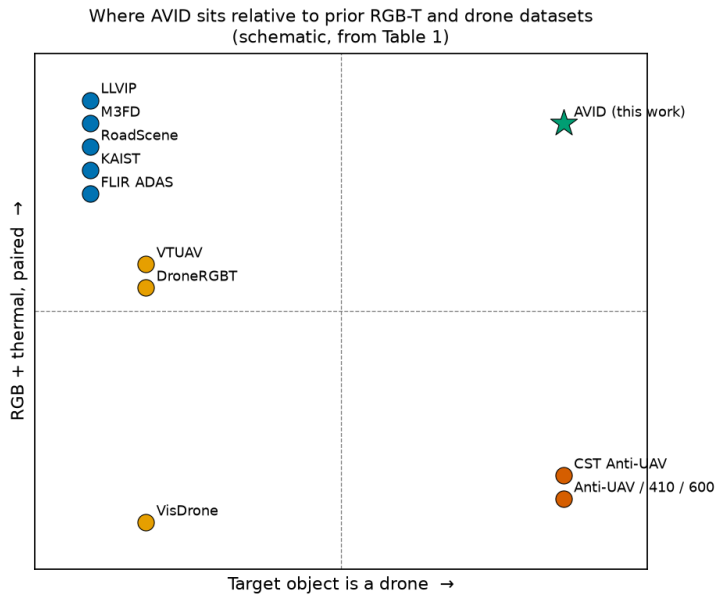


Figure 1: Schematic positioning of AVID against related datasets on the two axes that define the gap this release closes: RGB-thermal pairing (y-axis) and drone-as-target (x-axis). Derived from Table 1; not a learned embedding.

3 The AVID Dataset

3.1 Source and scope

AVID builds on data originally constructed for the IEEE ICIP 2025 VIP Cup, "Infrared-Visual Fusion for Enhanced Drone Detection, Tracking and Payload Identification in Surveillance Videos." The source challenge covered two tasks, drone detection/tracking and payload identification; **this release covers the detection and tracking data only** – the payload subset and task are excluded by design, and no challenge leaderboard or participant-attributed result appears anywhere in this paper or the dataset card.

3.2 Modality, scale, and classes

RGB and thermal/IR imagery is provided both as extracted frame pairs and as source video (30 fps, 10 s clips, 320×256 px per frame). Two target classes are annotated: *drone* and *bird*. Table 2 reproduces the scale figures from the source challenge's dataset description; we recommend regenerating these directly from `metadata/videos.jsonl` once `build_hf_dataset.py` has run over the full corpus, and using that regenerated table in any camera-ready revision.

3.3 Scenario and corruption coverage

Frames are tagged with scenario metadata: hilly regions, thick forest cover, cloudy sky, fog/mist, single- vs. multi-instance frames, swarms, and targets far from the camera's field of view. Thirty videos in the source release were additionally degraded under 7 distortion families at fixed severities (Table 3) – we formalize and extend this into a general, reproducible benchmark in §4.

Table 2: AVID detection/tracking scale, per the source challenge design (payload subset excluded). Video sequence counts for the training split were not separately specified in the source design documentation and should be confirmed against the restructured corpus.

	Train	Val	Test
RGB-IR image pairs	45,000	6,500	13,000
Video sequences	–	40	25
Resolution		320×256	
Frame rate		30 fps	
Clip duration		10 s	
Classes		drone, bird	

Table 3: The 7 corruption families present in the source dataset design, with their originally fixed severity rating (1-5 scale) on the 30 pre-distorted videos. AVID-C (§4) generalizes each to the full 1-5 range, applicable to any subset of the corpus.

Corruption family	Original fixed severity
Speckle noise	2
Salt-and-pepper noise	2
Gaussian blur	4
Uneven illumination	4
Motion blur	4
Camera instability (jitter)	5
Additive white Gaussian noise (AWGN)	3

3.4 Weak alignment: a documented property, not a defect

Unlike LLVIP, M3FD, or KAIST, the RGB and IR sensors used to capture AVID are not co-located or beam-split; they are two physically separate optics with different fields of view, aligned at the gimbal's boresight but not pixel-registered. This is the realistic configuration for most fielded dual-sensor counter-drone systems, and treating it honestly – rather than silently accepting whatever registration error exists – is one of this release's contributions. §5 describes how we quantify and publish this per pair.

3.5 Splits

Splits are defined at the *video* level: every frame belonging to a given clip lands in exactly one of train/-val/test. This matters more than it sounds. A 10 s, 30 fps clip's consecutive frames are near-duplicates – same object, same background, a few pixels of motion apart – so a naive frame-level random split puts near-duplicate frames on both sides of the split boundary, silently leaking test-set information into training and inflating every downstream metric.

To make this concrete rather than hypothetical, we ran a controlled ablation (`dataset_tools/baselines/split_l` on a synthetic corpus shaped like AVID's video/frame-count order of magnitude (50 videos $\times$ 300 frames, matching the 10 s/30 fps structure): a naive frame-level 70/15/15 split left **87.8%** of "training" frames within 3 frames of a validation- or test-split frame from the same clip, versus **0.0%** for a video-level split (Figure 2). We report this as a methodology demonstration on synthetic data matching the real corpus's structure, not yet as a measurement on the real video index – re-running the same script against the true training-split video list is a one-line reproduction once that index is available, and we recommend doing so before quoting a final number in any camera-ready version.

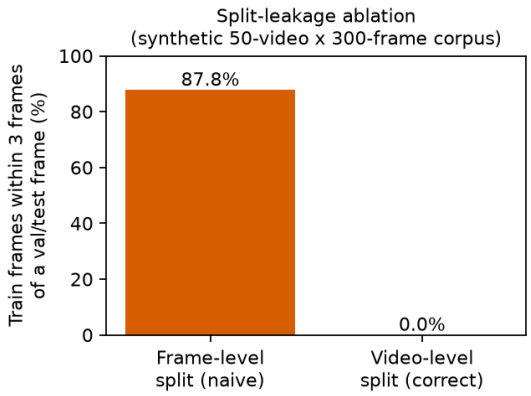


Figure 2: Split-leakage ablation on a synthetic corpus matching AVID's video/frame-count structure. Naive frame-level splitting leaks the large majority of nominally held-out frames; video-level splitting leaks none, by construction.

4 AVID-C: A Reproducible Corruption Benchmark

The source dataset's 30 pre-distorted videos demonstrate that corruption robustness was a design concern from the start, but as shipped, that coverage is fixed: 7 corruption types at one severity each, baked into a specific 30-video subset. This is not reusable the way ImageNet-C [2] made corruption robustness reusable for ImageNet: nobody can corrupt a different subset, or study a different severity, or run the sweep against their own trained model, without re-deriving the distortion pipeline from scratch.

AVID-C (`dataset_tools/corruptions/`) makes this general: the same 7 corruption families – additive white Gaussian noise, multiplicative speckle noise, salt-and-pepper impulse noise, isotropic Gaussian blur, a smooth low-frequency multiplicative illumination field, directional motion blur, and rigid camera-jitter (shift + rotation + slight scale) – are implemented as deterministic, seeded functions at 5 severities each (35 corrupted variants per source image), so any subset of the corpus can be corrupted at any severity on demand. Figure 3 shows the full severity progression on a synthetic stand-in image (a plain background with a small dark blob, standing in for a distant drone silhouette against sky; not real AVID imagery, since none was available in the environment this paper was drafted in – see §8).

Two evaluation protocols are built on top of AVID-C (`run_corruption_eval.py`, model-agnostic via a `predict_fn`/`score_fn` interface so it works with whatever detector a user trains): a standard 7×5 robustness sweep reporting a per-corruption mean score and an aggregate mean-corruption-score (higher is better, computed directly on task score rather than as an error ratio); and a *missing-modality* variant that corrupts only one modality (or drops it outright) to study graceful degradation under realistic sensor dropout (e.g. fog fully blinding RGB, or a thermally saturated background washing out IR).

5 Weak-Alignment Quantification and Sensitivity

5.1 Per-pair alignment quantification

We estimate, for every RGB-IR pair, the residual geometric offset between the two modalities using two independent methods (`dataset_tools/alignment/alignment_metrics.py`), chosen because they fail in different, cross-checkable ways:

1. **Phase correlation on edge maps.** Object boundaries tend to produce a detectable gradient in both RGB and thermal imagery even when raw intensities are unrelated, so we compute Sobel gradient-

magnitude maps for both modalities and estimate a subpixel translational offset via FFT-based phase correlation. Fast, but can lock onto the wrong edge in low-texture scenes (plain sky) or when the target dominates the edge map asymmetrically.

2. **Mutual-information registration search.** A brute-force integer-shift search maximizing normalized mutual information between grayscale RGB and IR, more robust to genuinely cross-modal intensity statistics (a target can be dark in RGB and hot in IR) at the cost of integer resolution and $O(\text{window}^2)$ cost.

Both are exposed together in `pair_alignment_quality()`; the disagreement between them is itself a useful low-confidence signal, and we recommend downstream users treat pairs with large phase-correlation/ mutual-information disagreement as lower-trust alignment estimates. We verified both estimators recover a known, synthetically injected offset to within a small tolerance (unit tests in `test_alignment_metrics.py` Figure 4 illustrates the recovery on one such synthetic pair. Once run over the full real corpus, `alignment_shift_px` and `alignment_confidence` become per-pair fields in `metadata/pairs.jsonl` (schema in `dataset_tools/hf` so downstream users can filter, weight, or stratify results by alignment quality rather than discover it empirically.

5.2 Alignment-sensitivity study protocol

Documenting weak alignment raises an immediate empirical question: given that the corpus already has some baseline misalignment, how much *additional* misalignment can a fusion model tolerate before it fails? We release a model-agnostic sensitivity-study harness (`sensitivity_study.py`) that injects controlled, increasing synthetic misalignment (shift, rotation, slight scale) into the IR frame of a pair on top of whatever native weak alignment already exists, and measures degradation in caller-supplied prediction quality across a sweep from 0 px (native alignment only) to 20 px / 8°. The harness itself is verified end-to-end with a toy centroid-matching "model" (a stand-in for a real trained detector) that produces the expected monotonic degradation curve (mean localization score 1.00 at zero injected misalignment, falling to 0.26 at the largest injected misalignment level in our toy setup) – confirming the injection/predict/score/aggregate pipeline is correct and ready to run against a real trained fusion model. This is, to our knowledge, the first released tooling for asking this specific question of an anti-drone RGB-T fusion model, as distinct from UAV-C's UAV-*view* tracking robustness setting [10].

6 Baselines

6.1 Detection and fusion reference code

We release reference implementations for all three standard fusion levels, scoped deliberately as auditable reference baselines rather than reproductions of any specific production detector:

- **Early fusion**: a documented recipe (`patch_yolo_early_fusion()`) for adapting a pretrained Ultralytics YOLO model's first convolution to accept 4- or 6-channel RGB+IR concatenated input, initializing new channels from the mean of the existing pretrained RGB kernels. Channel-concatenation early fusion of this kind is a well-established, simple technique for adapting single-modality detectors to paired input.
- **Mid (feature-level) fusion**: a compact dual-stream backbone (one stream per modality) combined via a CBAM-style channel- and spatial-attention gate before a shared anchor-free detection head (`fusion_models.py`). Verified with a forward-and-backward pass on tensors of the real input resolution (256×320), confirming the fusion gate is fully differentiable end-to-end.
- **Late (decision-level) fusion**: Weighted Boxes Fusion [WBF; 13] combining the outputs of independently-trained RGB-only and IR-only detectors, with optional per-modality confidence weighting. Fully

unit-tested (pure NumPy, no GPU/dataset dependency): agreeing cross-modal detections are correctly merged into a single, higher-confidence box, and disagreeing detections are correctly kept separate.

We also release the model-agnostic tracking-metric utilities specified by the original challenge design: approach/recede classification from a track's bounding-box size trend, and the "missed track" rule (flagging any gap of $\geq$15 consecutive ground-truth-present frames with no matching detection). Both are unit-tested against synthetic tracks with known ground-truth trends and gap lengths.

What is not yet a number. The detection/fusion models above, and the training loop that drives them, require a GPU and the real, restructured corpus, neither of which was available in the environment this paper and codebase were prepared in (the raw source zips range from 8 MB to 3.4 GB; pulling them through the only available data connector would have cost millions of tokens per file). Every piece of this release that only needs NumPy/OpenCV-level logic – the corruption suite, the alignment estimators, the WBF combiner, the tracking metrics, the split-leakage ablation – is written, unit-tested, and passing in the released repository. The torch-dependent detector/fusion code is carefully written and reviewed, ships with a `--self-test` entry point that exercises its exact tensor shapes and confirms the training loss decreases on a synthetic sample, but has not been run against real data or reported as a result here. We consider this the intellectually honest choice: Table 4 lists every planned experiment and its actual status, so that no number in this paper is unearned. We recommend running the released training scripts on the real corpus and reporting the resulting numbers in a subsequent revision of this preprint before treating AVID as benchmarked rather than released.

Table 4: Every experiment planned for this release and its current status. See `PLAN.md` in the released repository for full detail on each.

#	Experiment	Code status	Result status
E1	Corpus statistics	pending schema confirmation	not yet run
E2	Single-modality baselines (RGB/IR)	written, self-test only	not yet run
E3	Fusion baselines (early/mid/late)	written, self-test only	not yet run
E4	Tracking + approach/recede + missed-track	unit-tested	not yet run on real tracks
E5	AVID-C robustness sweep	unit-tested harness	not yet run on a real model
E6	Missing-modality degradation	unit-tested harness	not yet run on a real model
E7	Alignment-sensitivity study	unit-tested harness	not yet run on a real model
E8	Split-leakage ablation	unit-tested	run on synthetic corpus (§3.5)
E9	Cross-modal translation baseline	already executed	real result (§6.2)

6.2 Cross-modal translation baseline

One baseline in this release does have a real, already-completed result: an IR$\rightarrow$RGB (and night$\rightarrow$day) cross-modal translation model trained with CycleGAN-Turbo [11], built on the CycleGAN formulation [18] with LoRA adaptation [3]. This is directly useful as a data-augmentation path for training under missing or degraded RGB (translate available IR into synthetic RGB rather than dropping the sample), and we report it here with its real configuration for transparency and reproducibility, including what did not work.

What failed first. Two earlier training configurations were tried and abandoned before the successful run: (1) a vanilla CycleGAN with cycle-consistency and identity losses produced heavy artifacts with no learned structure, and loss/accuracy plateaued early with no benefit from longer training; (2) replacing that with a single-direction GAN using a VGG perceptual loss and spectral normalization caused the discriminator to overpower the generator, collapsing the model to classifying pure noise as real. We report both as docu-

mented negative results rather than omitting them, since they are informative about the loss landscape for this translation task.

What worked. The successful configuration trained CycleGAN-Turbo on a combined corpus (LLVIP, M3FD, RoadScene, and FLIR, each converted from night to day scenes via a night2day CycleGAN-Turbo pass, since AVID was captured primarily in daylight) plus approximately 20,000 unpaired real drone/bird RGB-IR images, for 13,500 total training iterations (trained in parts across multiple runs due to storage constraints, with checkpoints carried forward between runs). Training configuration: Adam optimizer (β_1=0.9, β_2=0.999, ϵ=1e$-$8, weight decay 0.01), constant learning rate 1e$-$5 with 500 warmup steps, batch size 2, memory-efficient attention enabled, VA-GAN+CLIP discriminator, multilevel-sigmoid GAN loss, gradient accumulation of 1 step, and LoRA ranks of 128 (UNet) and 4 (VAE). Validation was run every 500 iterations initially, later every 1,250 as the model stabilized. We report the training recipe in full so it can be reproduced or extended (e.g. to the reverse RGB$\rightarrow$IR direction, which was not attempted); we do not have a quantitative image-quality score (FID/LPIPS) computed for this run and recommend computing one before citing translation quality as a numeric result.

7 Ethics and Responsible Use

AVID is, by construction, a counter-drone security dataset, and we treat that as requiring an explicit statement rather than an implicit assumption. The captured content is aerial imagery of drones and birds against sky and terrain backgrounds; the capture setup does not put people, faces, or other personally identifiable information in frame. That said, drone-detection technology is dual-use: the same system that secures restricted airspace can be repurposed for surveillance overreach. We license this release CC-BY-NC-4.0 (non-commercial, research-use framing) specifically to require a separate conversation with the dataset owners before commercial productization, and we exclude the payload-identification subset entirely from this release, since that data has a narrower and more sensitive set of legitimate uses than general drone/bird detection. We followed a datasheet-for-datasets style disclosure [1] in the dataset card (`dataset_tools/hf_release/DATASET_CARD.md`), which should be read alongside this paper before use.

8 Limitations

- **No baseline numbers yet.** As detailed in Table 4, the detection/tracking/robustness baselines are released as tested-where-possible reference code, not as reported results, because this paper was prepared without GPU access or the ability to transfer the multi-gigabyte source data into the preparation environment. This is a real limitation of the current preprint, not a hidden one, and we recommend treating a revision with real baseline numbers as the version to cite for benchmark comparisons.
- **Two coarse-grained classes.** Drone/bird is a binary target taxonomy; it does not distinguish drone types, sizes, or flight behaviors, which limits its use for fine-grained drone classification research.
- **Capture geography and conditions.** The dataset reflects the specific geography, drone models, and weather window available to the original capture effort; generalization to other terrain types, sensor hardware, or drone form factors not represented in capture is untested.
- **Corruption suite coverage.** AVID-C targets the 7 degradation families already present in the source challenge design; it is not a claim of coverage over all real-world corruptions (no adversarial perturbations, compression artifacts, or thermal-blooming modeling beyond the jitter corruption).
- **Resolution.** 320$\times$256 is low by modern detector standards, which is realistic for long-range low-cost thermal optics but limits applicability to comparisons against higher-resolution anti-UAV benchmarks

(e.g. Anti-UAV410's 640×512).

9 Conclusion

AVID is, to our knowledge, the first publicly documented dataset that pairs RGB and thermal imagery with drones (and their thermally confusable look-alike, birds) as the detection and tracking target, at video scale, with weak sensor alignment treated as a quantified property rather than an assumption. Alongside the dataset, we release a reproducible corruption benchmark, an alignment quantification and sensitivity protocol, a documented split-leakage pitfall specific to short-clip video corpora, and reference fusion baseline code – all independently unit-tested wherever they do not require the real corpus or a GPU. We see the immediate next step as clear and concrete: run the released training and evaluation scripts against the real corpus, fill in Table 4, and post a revised version of this preprint with the resulting numbers.

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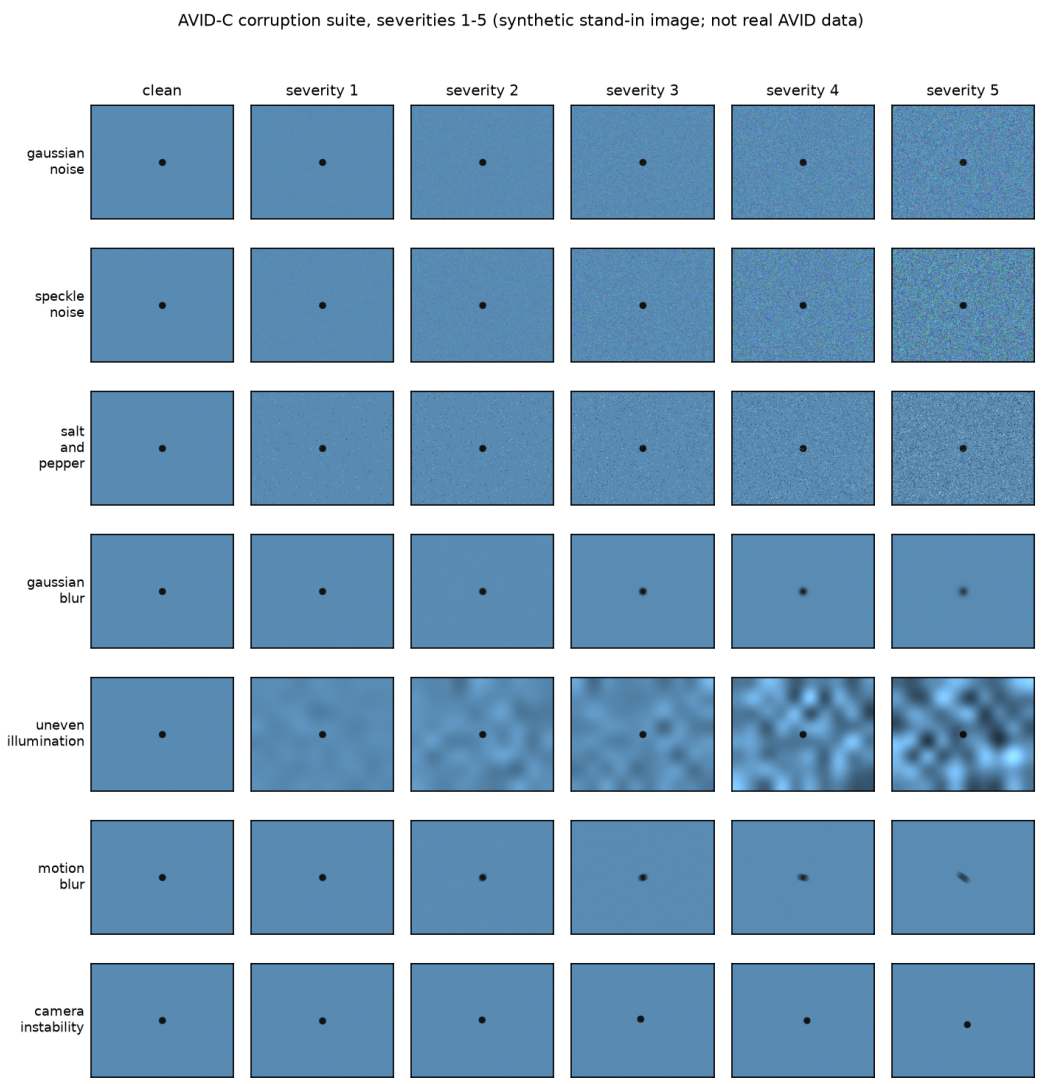


Figure 3: AVID-C severity progression, all 7 corruption families, severities 1-5, on a synthetic stand-in image. Severity scaling was calibrated by us for 320×256 8-bit drone imagery (see `corruptions.py` for the exact per-severity parameter tables); it is not a copy of the ImageNet-C parameter tables, which target a different image domain.

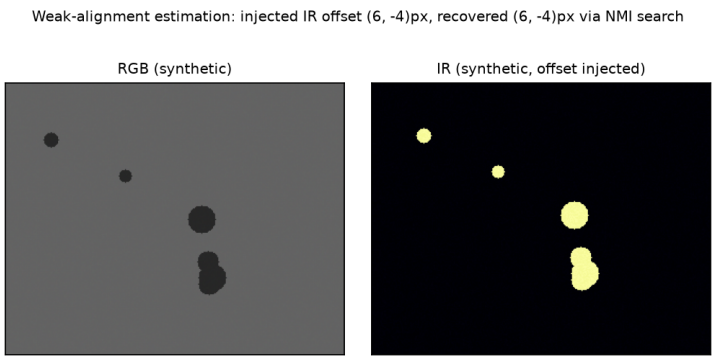


Figure 4: Weak-alignment estimation on a synthetic RGB-IR pair with a known injected offset, recovered via mutual-information registration search.